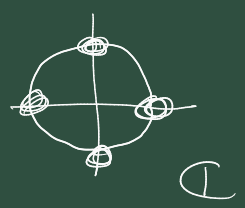
Isomorphisms

Consider $\mathbb{Z}_4 = \{0, 1, 2, 3\}$ under addition modulo 4.

Subgroup $H = \langle i \rangle$, of $\mathbb{T}$,

$\mathbb{T} = \{ z \in \mathbb{C} : |z| = 1 \}$ under complex multiplication.

$H = \langle i \rangle = \{1, i, -1, -i\}$



Both $\mathbb{Z}_4, H$ are of order 4.

Consider their Cayley tables.

$\mathbb{Z}_4, +$	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

$H, \times$	1	i	-1	$-i$
1	1	i	-1	$-i$
i	i	-1	$-i$	1
-1	-1	$-i$	1	i
$-i$	$-i$	1	i	-1

How similar are the Cayley tables? "The same" This is an example of an isomorphism between the two groups.

Def Isomorphism

Two groups $(G, \cdot)$, $(H, \circ)$ are <u>Isomorphic</u> means

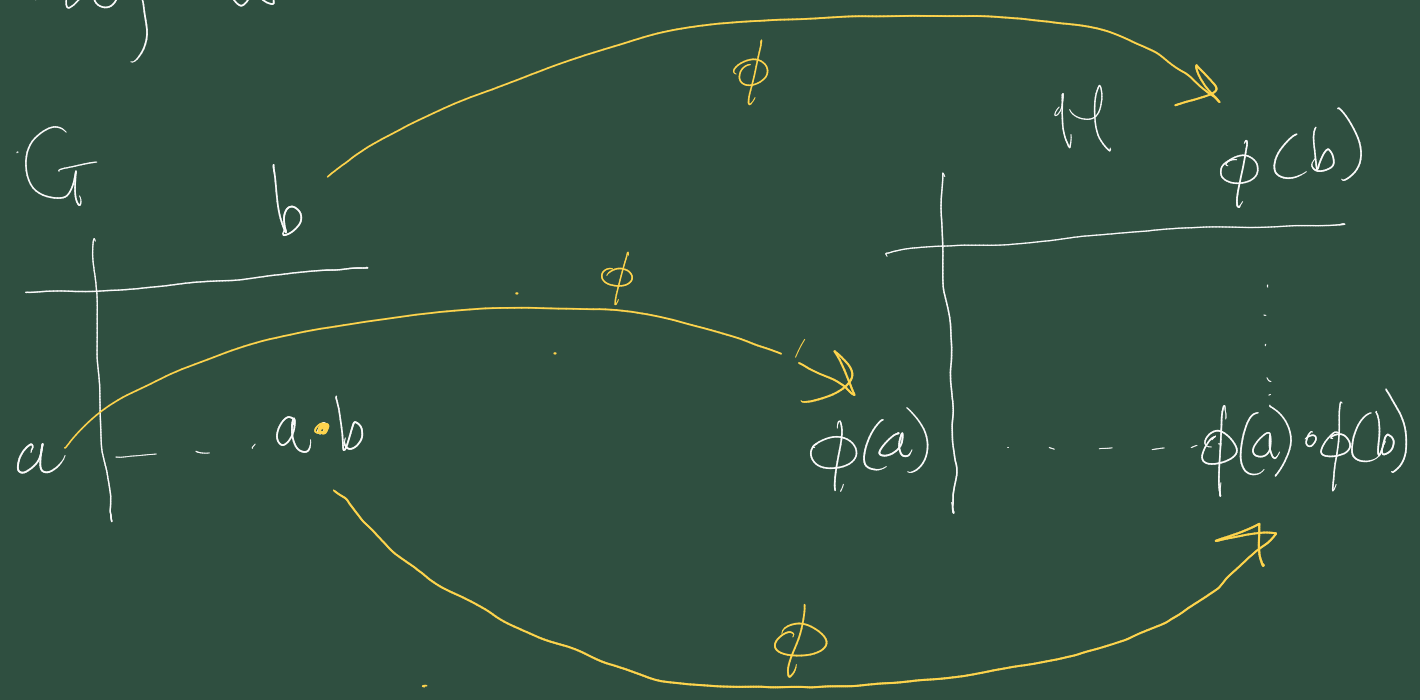
$\exists$ an bijective mapping

$$\phi : G \longrightarrow H$$

satisfying the "homomorphism property"

$$\forall a, b \in G \quad \phi(a \cdot b) = \phi(a) \circ \phi(b)$$

If so, we write $G \cong H$



Eg. In ~~[illegible]~~ previous example.

$$\phi : \mathbb{Z}_4 \longrightarrow H.$$

$$0 \longmapsto 1$$
$$1 \longmapsto i$$
$$2 \longmapsto -1$$
$$3 \longmapsto -i$$

and the fact that bodies of the tables end up with "the same" coloring means the homomorphism property is satisfied.

<u>Ex 9.2</u> $\phi : \mathbb{R} \longrightarrow \mathbb{R}^+$

$$\begin{tikzcd} \mathbb{R} \arrow[d] & \mathbb{R}^+ \arrow[d] \\ (\mathbb{R}, +) & (\mathbb{R}^+, \times) \end{tikzcd}$$

$$\mathbb{R}^+ = \{x \in \mathbb{R} : x > 0\}.$$

Claim is the mapping $\exp : \mathbb{R} \longrightarrow \mathbb{R}^+$

$$x \longmapsto e^x$$

is an isomorphism.

1) exp is a bijection $\mathbb{R} \to \mathbb{R}^+$, with inverse $\ln$, the logarithm.

2) Homomorphism property is our well known property of indices / exponents.

$$\boxed{\exp(x+y)} = \boxed{e^{x+y} = e^x e^y}$$

$$= \exp(x) \times \exp(y)$$

<u>Eg 9.3</u> Consider $(\mathbb{Z}, +)$,

subgroup of $\mathbb{Q}^* = \{q \in \mathbb{Q} : q \neq 0\}$

$$\langle 2 \rangle = \{ 2^n : n \in \mathbb{Z} \}$$

and the mapping

$$\phi : \mathbb{Z} \longrightarrow \langle 2 \rangle$$

$$m \longmapsto 2^m$$

Again, the homomorphism property is the well known rule of exponents.

$$\phi(m+n) = 2^{m+n} = 2^m 2^n = \phi(m)\phi(n)$$

Eg 9.4 Clearly $\mathbb{Z}_8$ and $\mathbb{Z}_{12}$ are not isomorphic since they are of different sizes $|\mathbb{Z}_8| = 8$ $|\mathbb{Z}_{12}| = 12$ i.e. $\mathbb{Z}_8 \not\cong \mathbb{Z}_{12}$

However

$$U(8) = \{1, 3, 5, 7\}$$

$$U(12) = \{1, 5, 7, 11\}$$

these might be isomorphic.

$U(8)$

	1	3	5	7
1	1	3	5	7
3	3	1	7	5
5	5	7	1	3
7	7	5	3	1

$U(12)$

	1	5	7	11
1	1	5	7	11
5	5	1	11	7
7	7	11	1	5
11	11	7	5	1

From this we see that the mapping $\phi : U(8) \longrightarrow U(12)$

$$1 \longmapsto 1$$

$$3 \longmapsto 5$$

$5 \longmapsto 7$

$7 \longmapsto 11$

satisfies the homomorphism property.

and so $U(8) \not\cong U(12)$.

Either of these ~~isomorphic to $\mathbb{Z}_4$~~ ~~or H~~, from the first example.

$\mathbb{Z}_4, +$	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

$H, \times$	1	i	-1	$-i$
1	1	i	-1	$-i$
i	i	-1	$-i$	1
-1	-1	$-i$	1	i
$-i$	i	1	i	-1

We can see that the mappings defined by this coloring are not isomorphisms.

To be sure the $U(8) \not\cong \mathbb{Z}_4$ we have to consider all other bijections

Can show (exercise) that if $\phi : G \to H$ is an isomorphism then $\forall x \in G$ $|\phi(x)| = |x|$

So using this we can see (from tables) that $\mathbb{Z}_4 \not\cong U(8)$.

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Eg 9.5. Is $S_3 \cong \mathbb{Z}_6$?

S_3 the group of all permutations of $\{1,2,3\}$

$\mathbb{Z}_6$ integers under addition mod 6.

$|S_3| = 3! = 6$.

We know S_3 is non-abelian, whereas $\mathbb{Z}_6$ is abelian. This will mean $S_3 \not\cong \mathbb{Z}_6$.

Proof. Let $a, b \in S_3$ that do not commute.

ie. $a = (1\,2)$, $b = (1\,2\,3)$

$ab = (1\,2)(1\,2\,3) = (2\,3)$

$ba = (1\,2\,3)(1\,2) = (1\,3)$

so $\boxed{ab \neq ba.}$

But suppose $\phi : \mathbb{Z}_6 \to S_3$ is an

isomorphism and lets say.

$\phi(m) = a$, $\phi(n) = b$.

$$\boxed{ab} = \phi(m)\phi(n) = \phi(m+n) \text{ , hom. prop.}$$
$$= \phi(n+m) \text{ , } \mathbb{Z}_6 \text{ is abelian}$$
$$= \phi(n)\phi(m) \text{ , hom prop}$$
$$= \boxed{ba}.$$

This contradicts the property that a, b do <u>not</u> commute. So there can be no isomorphisms between $\mathbb{Z}_6$ to S_3, so $\mathbb{Z}_6 \not\cong S_3$.

Some initial theory about isomorphism.

<u>Theorem 9.6</u> Assume $\phi : G \to H$ is an isomorphism.

<u>Proofs.</u>

1. Claim: $\phi^{-1} : H \to G$ is also an isomorphism

<u>Proof</u> ϕ^{-1} exists, since ϕ is bijective.

Does ϕ^{-1} <u>satisfy the homomorphism property.</u>

Know $\boxed{\forall a, b \in G \quad \phi(ab) = \phi(a)\phi(b)}$.

we <u>want</u> <u>to</u> <u>prove</u> $\forall x, y \in H \quad \phi^{-1}(xy)$
$$= \phi^{-1}(x)\phi^{-1}(y)$$

Let $x, y \in H$. and let us write.

$a = \phi^{-1}(x)$, $b = \phi^{-1}(y)$, for some $a, b \in G$.

i.e. $\phi(a) = x$, $\phi(b) = y$.

$$\boxed{\phi^{-1}(xy)} = \phi^{-1}(\phi(a)\phi(b)),$$

$$= \phi^{-1}(\phi(ab)), \text{ hom. prop for } \phi$$

$$= ab, \text{ since } \phi, \phi^{-1} \text{ are inverses.}$$

$$= \phi^{-1}(x)\phi^{-1}(y).$$

2. Well for finite ~~group~~ sets, existence of an bijection between the sets implies they have the same size.

For infinite sets, existence of a bijection between sets is the <u>definition</u> of $|G| = |H|$.

3. This is essentially the idea behind our calculation in the $S_3, \mathbb{Z}_6$ example.

Assume G is abelian and $\phi : G \to H$ an iso.

Let $x, y \in H$ and let $\phi(a) = x$, $\phi(b) = y$, for some $a, b \in G$.

$$\boxed{xy} = \phi(a)\phi(b)$$

$$= \phi(ab), \text{ hom. prop for } \phi.$$

$$= \phi(ba), \quad G \text{ is abelian}$$

$$= \phi(b)\phi(a), \text{ hom prop for } \phi.$$

$$= yx.$$

Therefore H is abelian.

4. <u>Assume G is cyclic</u>, i.e. $G = \langle g \rangle$ for some $g \in G$. ($\phi : G \to H$).

<u>Claim</u> H is cyclic with generator $\underline{\phi(g)}$

<u>Proof</u> Let $x \in H$, we want to show

$$x = \phi(g)^n, \text{ for some } n \in \mathbb{Z}.$$

$$\begin{tikzcd} G \arrow[r] & H. \\ a \arrow[u, phantom, "\in"] & x \arrow[u, phantom, "\in"] \arrow[l, "\phi^{-1}"] \end{tikzcd}$$

Can write $a = \phi^{-1}(x)$, for some $a \in G$, (or vice versa $\phi(a) = x$). We know G is cyclic so $\underline{\exists\, n \in \mathbb{Z}}$ such that $\underline{a = g^n}$

$$\begin{aligned} x = \phi(a) &= \phi(g^n) \\ &= \phi(ggg \cdots g) \\ &= \phi(g)\phi(g)\phi(g) \cdots \phi(g), \\ &= \phi(g)^n. \end{aligned}$$

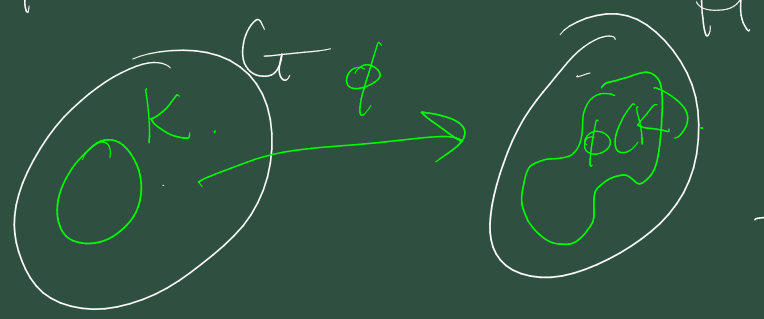
generalisation (through induction) of hom prop for ϕ.

This proves that $\langle \phi(g) \rangle = H$, and so H is cyclic too.

5.) Let K be a subgroup of G, $|K| = n$.

<u>Claim</u> $\underline{\phi(K)}$ is a subgroup of H of order n.

$\phi : G \longrightarrow H$.



<u>Proof</u>

1: $\phi(K) = \{\phi(k) : k \in K\}$ is a subset of H and $|\phi(K)| = |K| = n$ since ϕ is a bijection.

2. Is $\phi(K)$ a subgroup of H. Prove this using Prop 3.30.

1. Well $e_G \in K$.

Infact, can prove that $\phi(e_G) = e_H$.

Since given any $x \in H$

$x = \phi(a)$, for some $a \in G$

$= \phi(e_G a)$

$= \phi(e_G)\,\phi(a)$

$= \boxed{\phi(e_G)\,x}$.

Can also prove $x = x\,\phi(e_G)$

Therefore $e_H = \phi(e_G)$.

$\Longrightarrow$ $e_H \in \phi(K)$, since $e_G \in K$.

2. Let $x, y \in \phi(K)$, and $x = \phi(a)$, $y = \phi(b)$ for some $a, b \in K$.

$$\begin{aligned} xy &= \phi(a)\,\phi(b) \\ &= \phi(ab), \quad \text{hom. prop of } \phi. \\ &\in \phi(K), \quad \text{since } ab \in K \text{ as } K \text{ is a subgroup.} \end{aligned}$$

3. Similar idea.

<u>Claim</u> Given any $a \in G$,

$$\phi(a^{-1}) = \phi(a)^{-1}$$

Can prove this using hom-prop for ϕ (exercise).

So given $x \in \phi(K)$, with $\phi(a) = x$, $a \in K$.

$$x^{-1} = \phi(a)^{-1} = \phi(a^{-1}) \in \phi(K)$$

because $a^{-1} \in K$, as K is a subgroup

So by prop 3.30 $\phi(K)$ is also a subgroup of H.